GIUSEPPE BELLAMACINA

AI/ML ENGINEER

CONTACTS

Catania, Italy

+39 3496790342

bellamacina50@gmail.com

GitHub

Linkedin

Portfolio

LANGUAGES

- Italian (Native Speaker)
- English (B2)

PROGRAMMING LANGUAGES

- C
- C++
- C#
- Python
- Java
- JavaScript

QUERY LANGUAGES

- Cypher
- SQL
- SPARQL

GRAPHICS SOFTWARE

- Blender
- Photoshop

GAME DEVELOPMENT

- Unity
- Unreal

EDUCATION

Master's Degree in Computer Science

Università degli Studi di Catania | 2025 - Present

Bachelor's Degree in Computer Science

Università degli Studi di Catania | 2021 - 2024

- Graduation Grade: 110 cum Laude

WORK EXPERIENCES

AI Software Engineer

Ricca IT | September 2025 - Present

LLM Systems & Multi-Agent Architectures:

- Designed and deployed LLM-based systems for real-world applications, including solutions used by ASP (Azienda Sanitaria Provinciale), a regional public healthcare organization
- Built multi-agent architectures for complex reasoning tasks, separating planning, execution, and validation into coordinated components
- Developed end-to-end RAG pipelines (data ingestion, indexing, retrieval, reranking, generation) with focus on scalability and latency
- Integrated heterogeneous AI modules into cohesive, production-ready systems, improving reliability and maintainability

Model Adaptation & Evaluation:

- Applied model adaptation techniques including fine-tuning, reinforcement learning (GRPO), knowledge distillation, and domain adaptation
- Evaluated LLM systems beyond standard benchmarks, focusing on robustness, consistency, and behavior under real-world constraints

AI / Data Science Engineer

Intellisync | June 2024 - May 2025

LLM Systems & Applications:

- Developed LLM-based and multi-agent solutions for real-world applications, including systems used by the Italian Air Force Academy (Accademia Aeronautica Militare Italiana)
- Designed and implemented RAG architectures, from requirement analysis to system design and deployment
- Built agentic pipelines translating natural language into executable queries across multiple backends (SQL, graph databases via Cypher, semantic systems via SPARQL, and vector search)

Data Science & Predictive Systems:

- Developed predictive models on large-scale wind turbine datasets, including anomaly detection and energy loss forecasting
- Contributed to solutions later integrated into Yokogawa products through collaboration with BaxEnergy and Farsight
- Performed advanced data analysis to support operational decision-making